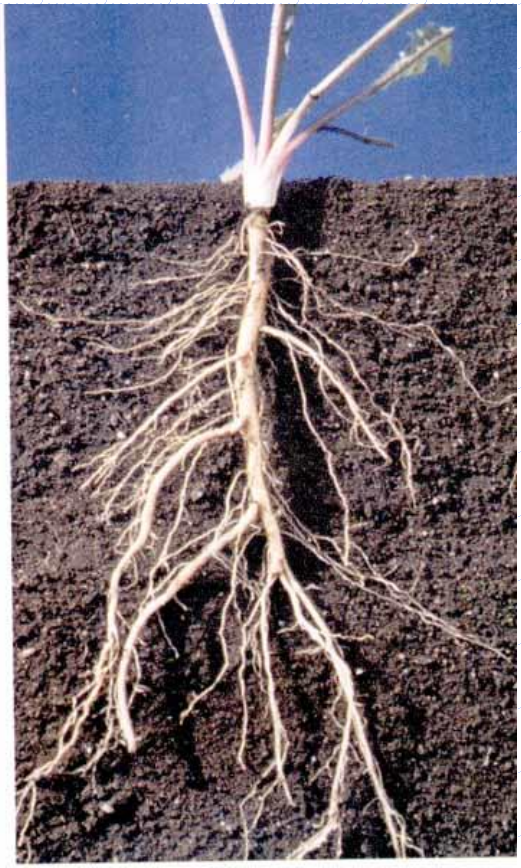
Root: Forms and Functions

Functions of a Root System

- 1. Absorb water and minerals from the environment
- 2. Anchor the plant in the ground
- 3. Store food that has been made in the leaves by photosynthesis
  - Can be used later by the plant to grow and survive



Taraxacum officinale – the common dandelion

What Are the Parts of a Root?

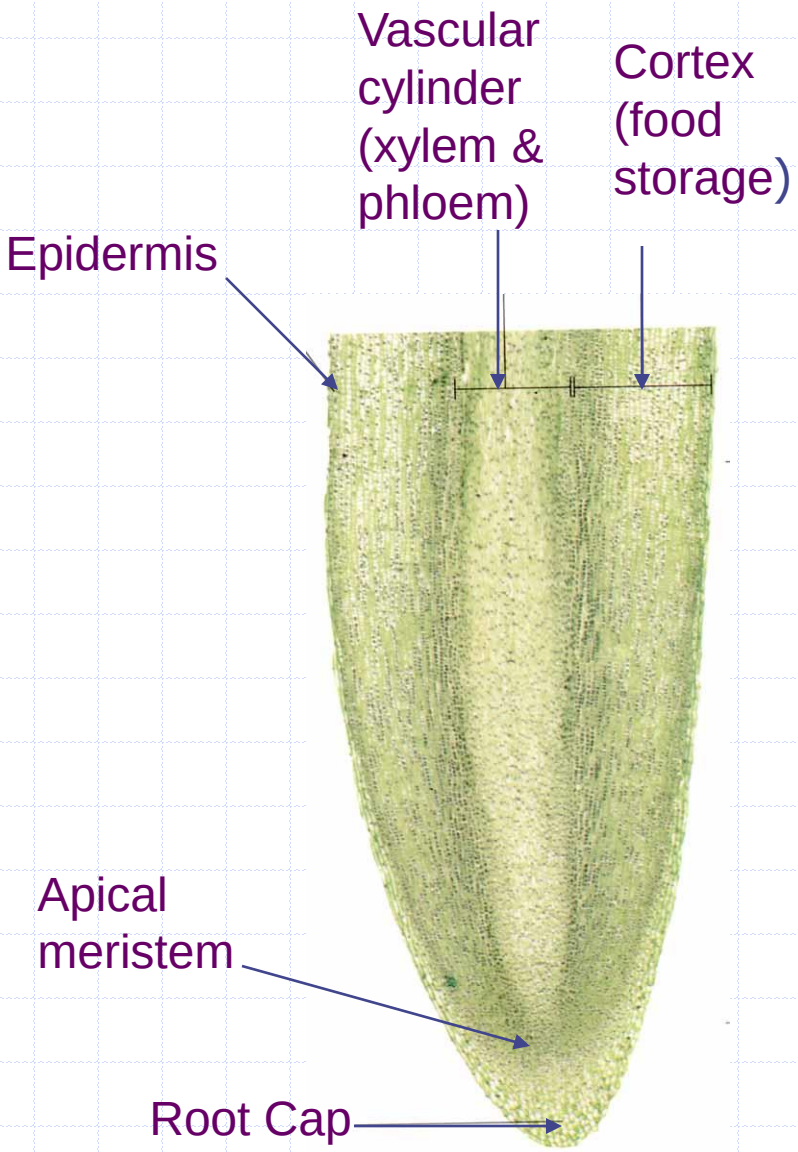


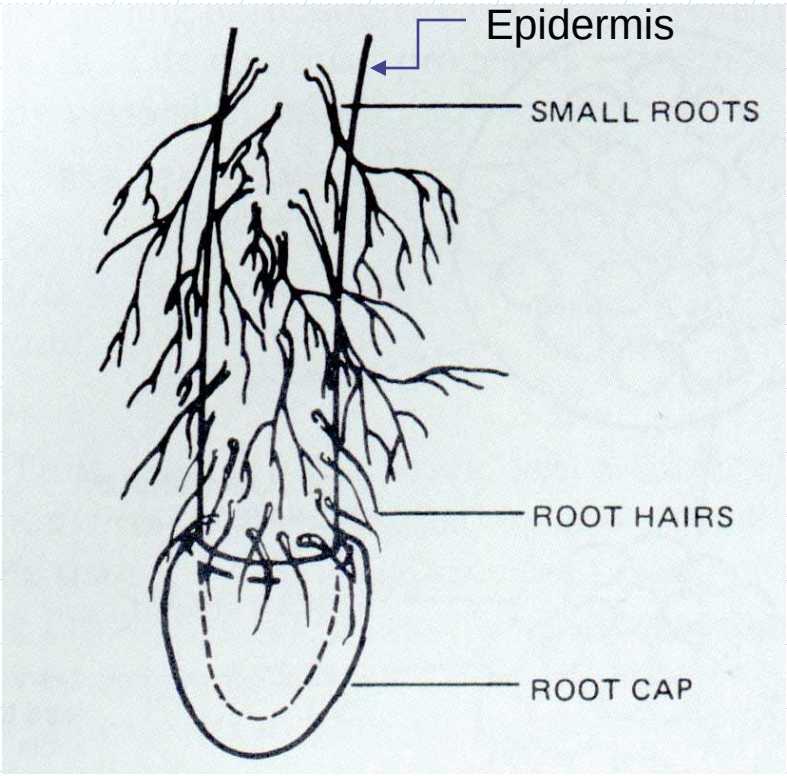
Cotyledon

Radicle = Primary root; notice all the root hairs

- When a seed germinates, the first structure to appear is the root, or radicle
  - A. It becomes the ***primary root***
    - Usually the most important root in some plants
  - B. Other roots branch out from the primary root; called ***secondary roots***

- C. The ***apical meristem***, found at the root tip, is where new cells develop
  - It is covered by the ***root cap*** – protects it from damage as it passes through coarse soil particles

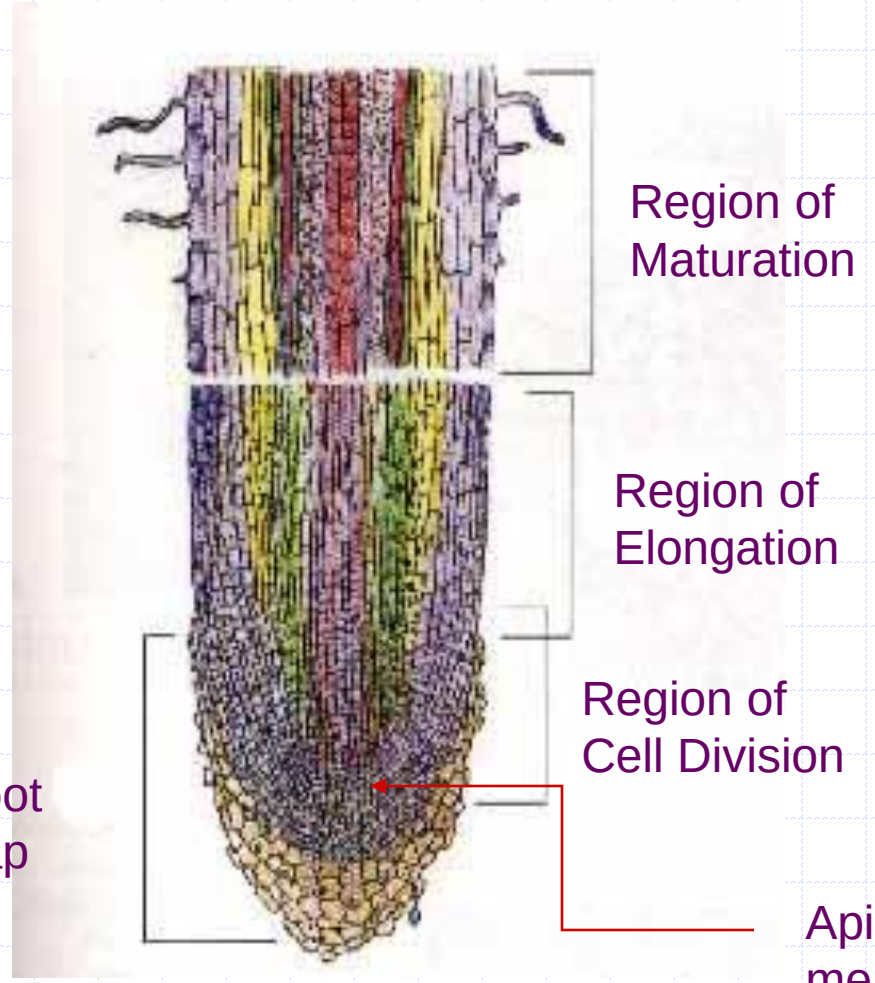




- D. The surface of the root is protected by skin cells called the ***epidermis***
  - Where water and minerals enter the root by osmosis & diffusion
  - Can grow long, hair like projections called ***root hairs***
    - They greatly increase the surface area of the root to allow more water intake

- There are three major zones of root development
  - 1. Region of Cell Division – this is where the cells start to divide through mitosis
    - Remember, this is where the apical meristem is located
  - 2. Region of Elongation – in this area just above the apical meristem, cells become longer and wider
    - This action helps to push the root tip through the soil
  - 3. Region of Maturation – at this point, the cells become specialized, or differentiated, taking on specific primary functions
    - For example, the development of root hairs, xylem, phloem, etc.

Regions of Cell Development

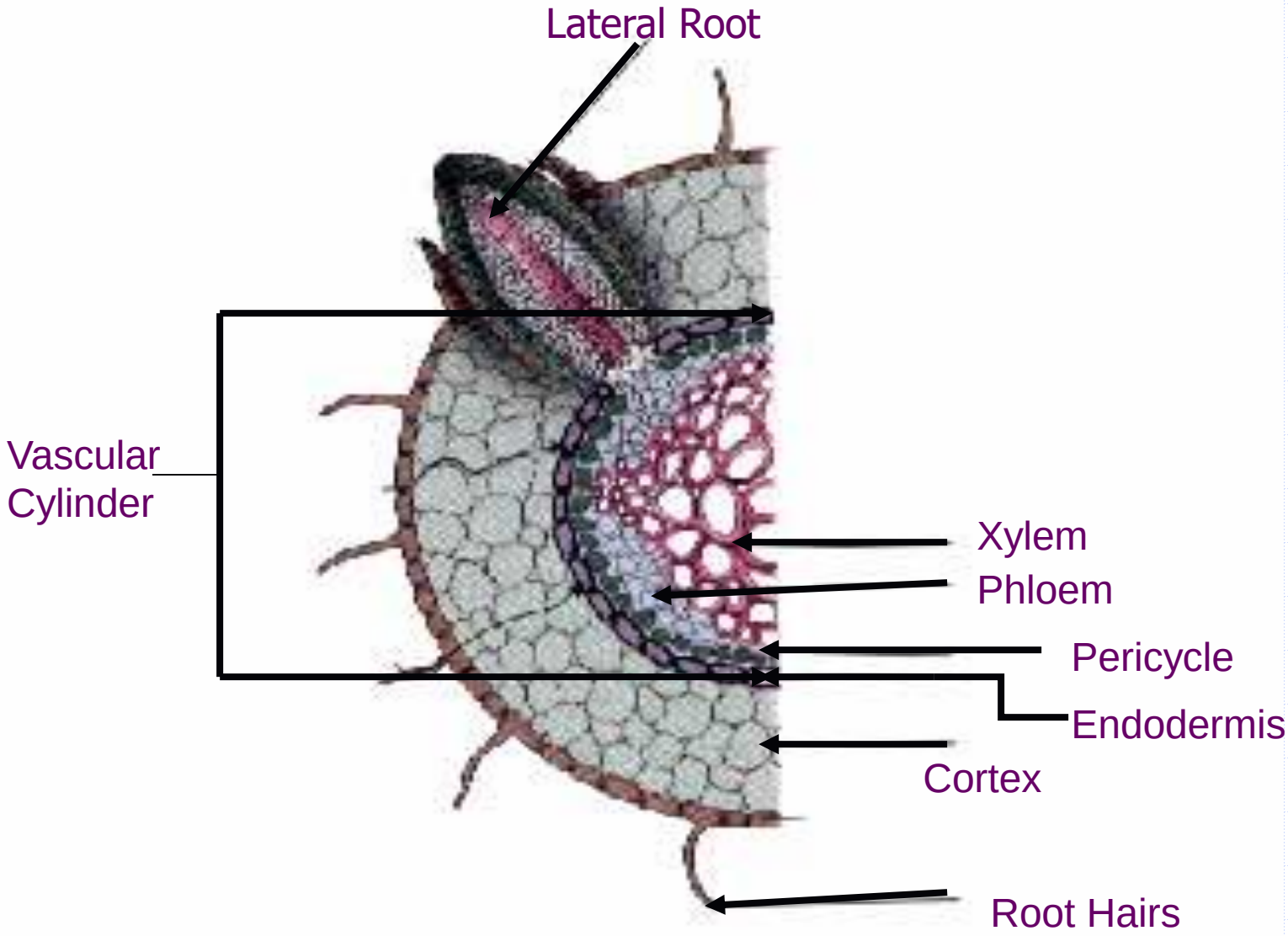


- Notice that cell division occurs at the tip of the root
- Older cells are found farther away from the root tip

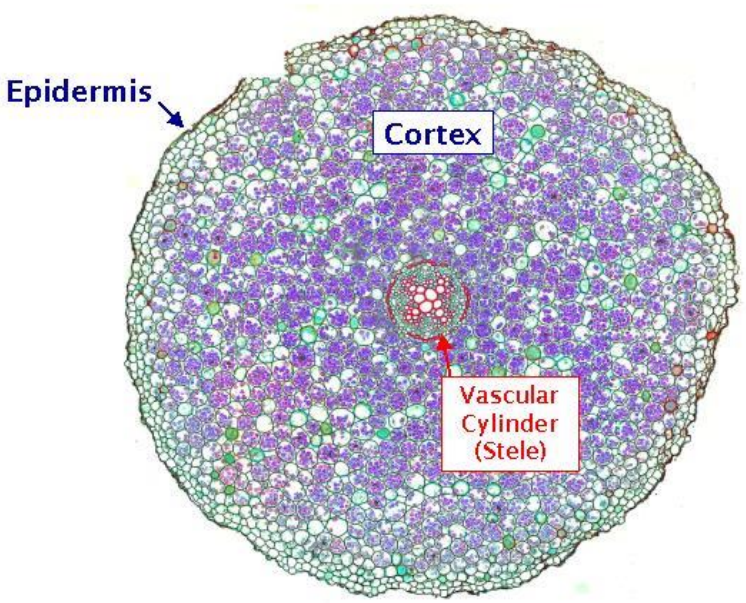
- There are several other layers of cells found within the root
  - A. Cortex – this is where the root stores the food produced by the leaves; it is made out of parenchyma tissue
  - B. Endodermis – the inner boundary of the cortex; one cell layer thick; controls the movement of water and minerals into the xylem and phloem
  - C. Pericycle – a layer found inside the endodermis which may produce lateral roots; also produces vascular cambium in dicots

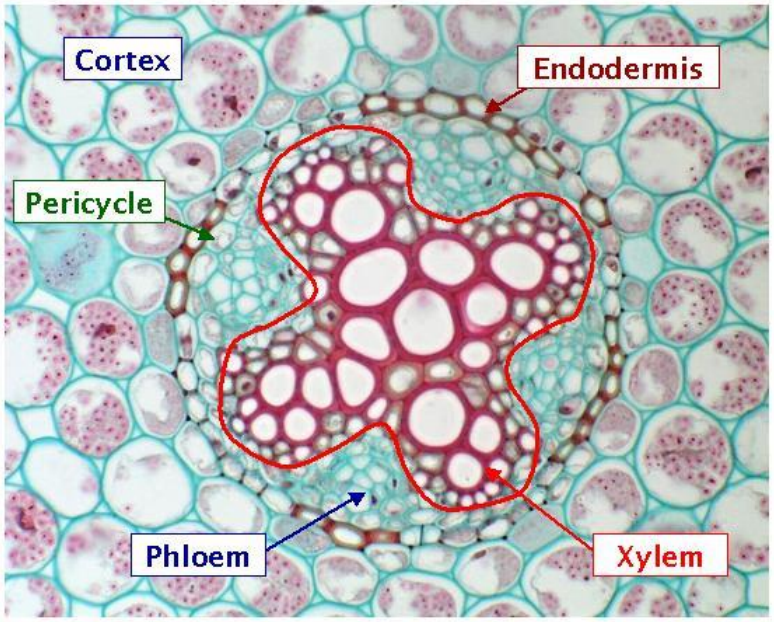
- D. Vascular cylinder, or stele – forms the central cylinder of the root; it is made out of two secondary transport tissues
  - 1. Xylem – carries the water taken in by the root hairs and epidermis up the plant to the stems and leaves
  - 2. Phloem – carries the food produced by the leaves down to the roots to be stored in the cortex

Internal Layers of the Root

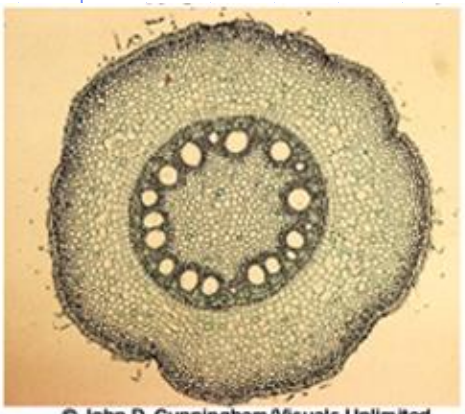


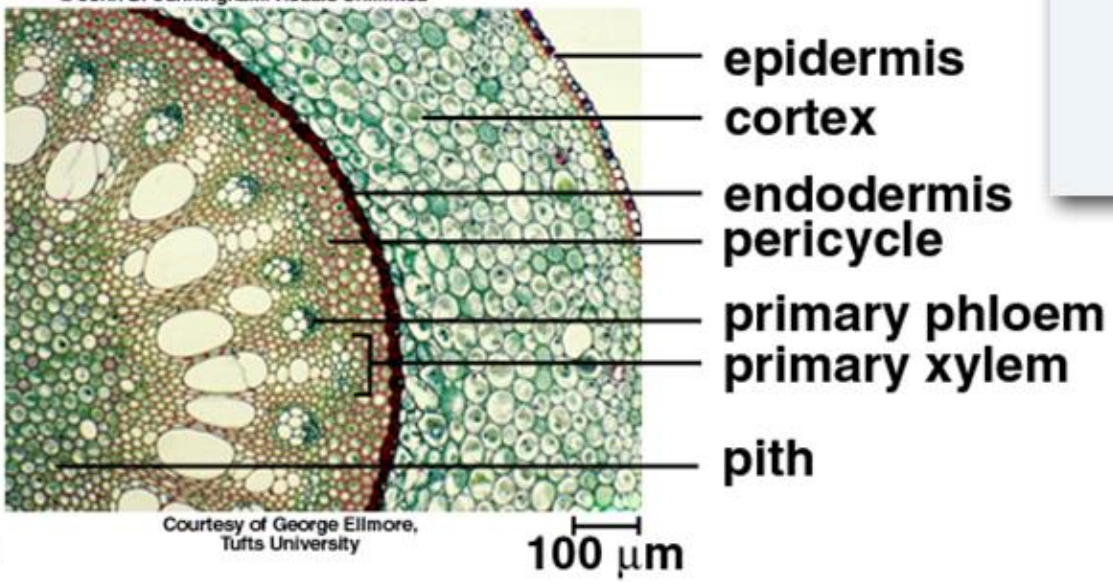
Dicot Root in Cross Section

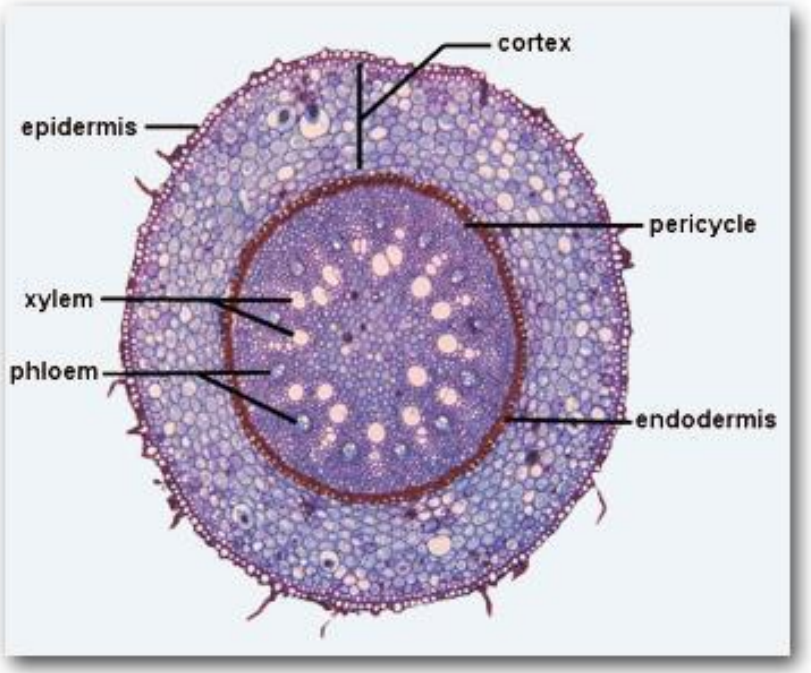




Monocot Root in Cross Section







What Are the Types of Root Systems?

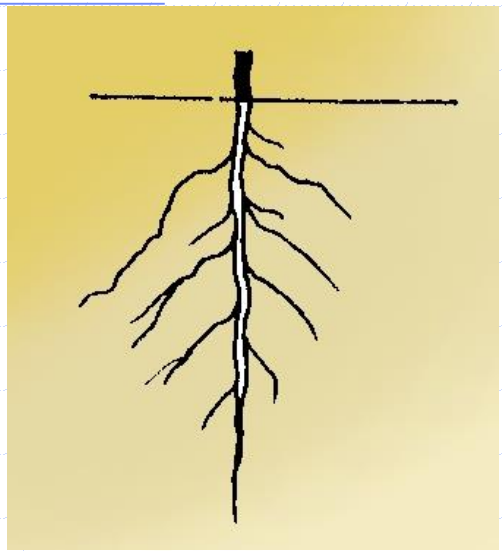
- Plant root systems are organized in two basic ways; It has to do with primary and secondary roots
  - A. A root system comprising one main primary root and many secondary roots branching off the primary root is called a ***taproot system***
    - Their roots reach far into the ground; they can be several feet long

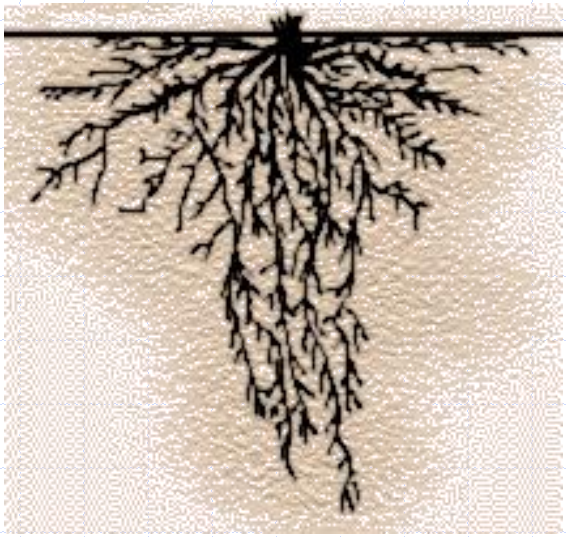


Fibrous root system of grass

- B. A system which has no dominant primary root but is made of many primary and secondary roots of similar size is called a ***fibrous root system***
  - The roots are smaller, shorter and more compact; They usually never grow below the first 6-12 inches of soil
    - These roots form a large network underground

Tap root and Fibrous (Diffuse) Root Systems – Both arise from radicle









- C. A system in which the roots arise from anything other than the radicle is called ***adventitious root system***.





Aerial roots are adventitious roots which develop from aerial parts of the plant and modified to perform various functions.

a). Mechanical Support

Prop roots (e.g. ***Ficus benghalensis***)

Adventitious roots arise from horizontal branches and grow vertically downwards. They function as pillars (**columnar roots**), giving mechanical support to the branches.

Stilt roots (e.g. ***Pandanus***)

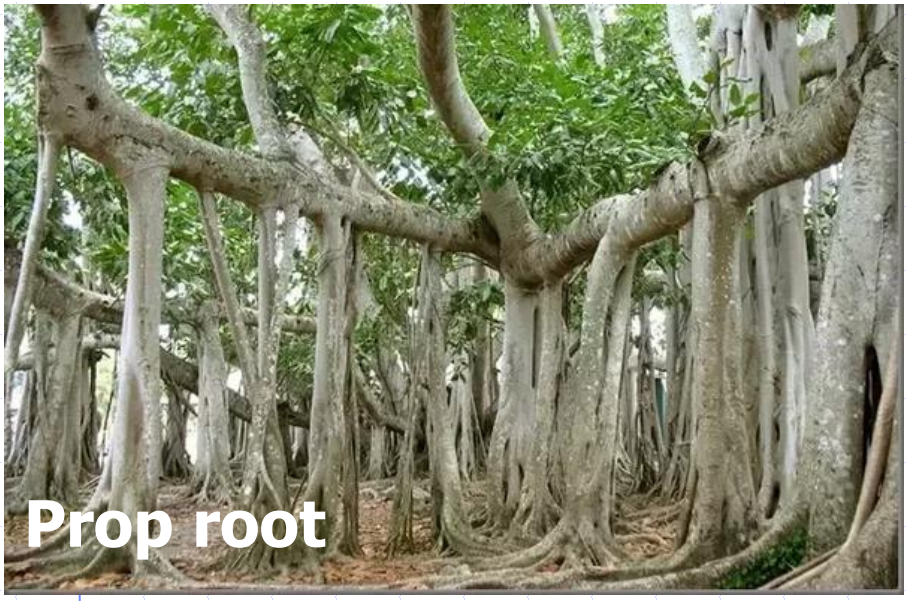
Adventitious roots arise from lower part of main stem and grow obliquely towards the soil. They keep the plant erect by giving additional support.

Climbing roots (e.g. ***Piper betel***)

Roots arise from nodes and help in attaching the climbing stem firmly to a tree or a wall, by various mechanisms.

Buttress roots **(**e.g. ***Prombab malabaricum****)*

These roots are helpful in fixing the plant at its normal position.


Prop root


Stilt root


Climbing root
Piper betle L.
Photo: Surajit Koley


Buttress root
shutterstock.com · 344009099

b). Sucking roots

Root penetrates into tissue of host plant and draw nutrients from the host. Such parasitic plants are not completely photosynthetic.



c). Respiratory roots (Pneumatophores)

Plants growing in marshy ,water-logged soils which contain almost no air, develop some branches which grow vertically upwards into the air. Such roots are provided towards the upper end with numerous pores through which gases diffuse in and out.



d). Floating roots

Roots usually develop above the level of water and serve to store up air and help in buoyancy of the plant. In addition, floating roots may be respiratory.



e). Epiphytic roots

Plants grow on other plants, and have special adventitious roots, the outer region of which is made up of velamen tissue. This tissue absorbs moisture from the air and makes it available to the plant.

